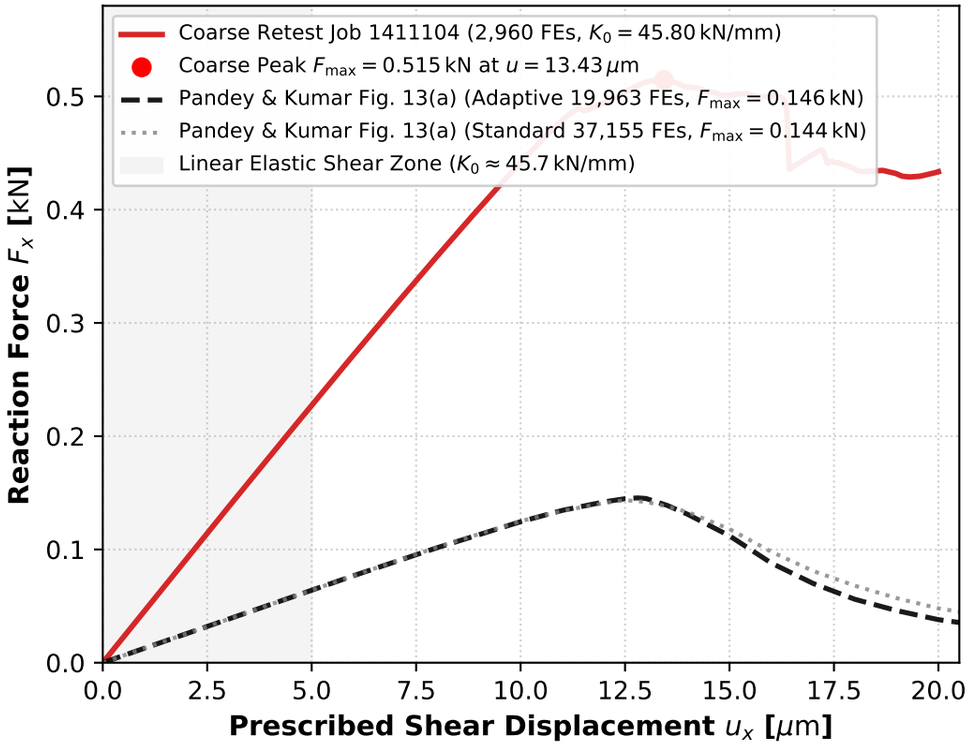
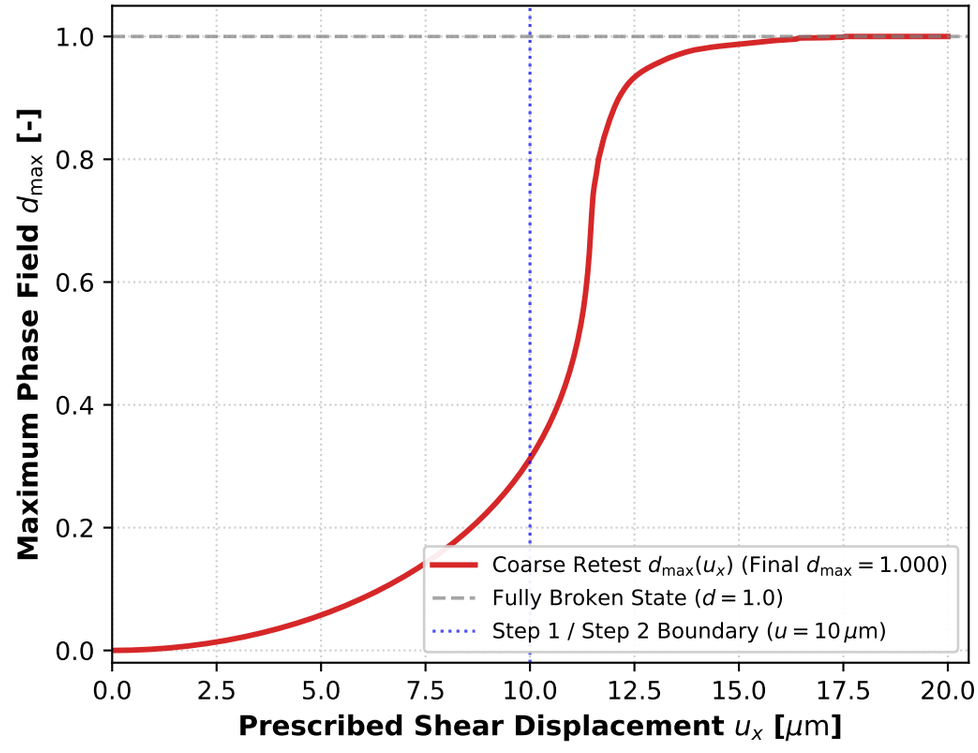
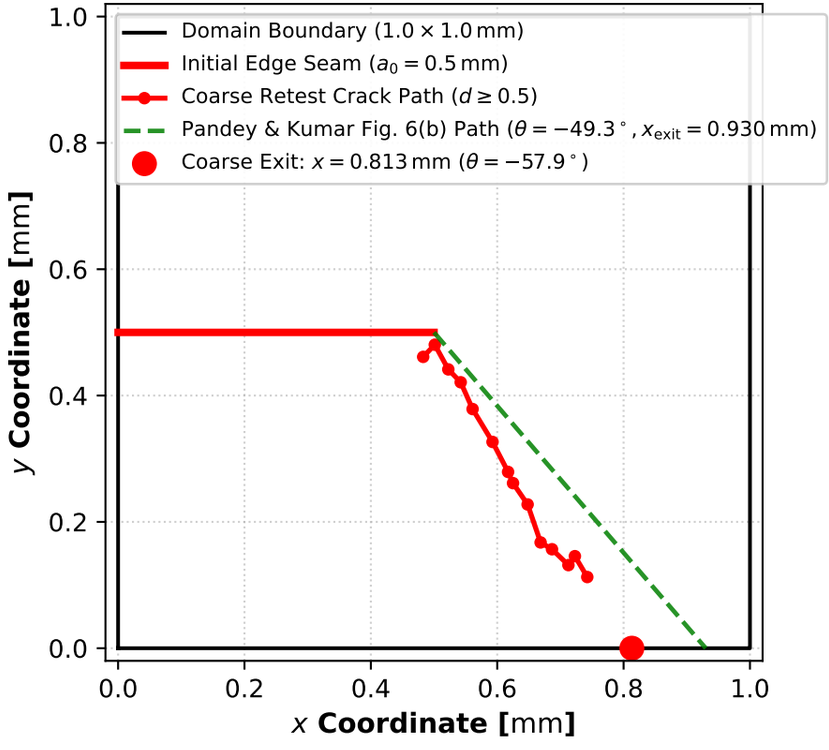
(a) Mode-II Global Force-Displacement Response



(b) Phase-Field Damage Growth & Full Horizon Saturation



(c) Computed Mode-II Crack Trajectory vs Literature Path



(d) Gate M2-4 Discretization Metrics & Formulation Verification

Quantity / Metric	Coarse Retest (J1)	Adapted Retest (J2	Literature Ref (Fig 13
Finite Element Coun	2,960 FEs	22,530 FEs	19,963 FEs (Adaptive)
Node Count	3,037 Nodes	22,642 Nodes	N/A (Published)
Initial Stiffness K0	45.80 kN/mm	45.49 kN/mm (In-Si	12.80 kN/mm (Apparent)
Reaction Force at 4.15	189.9 N	188.58 N (In-Situ)	~53.0 N
Peak Load F_max	514.51 N	Solving (Active)	145.5 N (19.9k FEs)
Displacement u(F_ma	13.43 um	Solving (Active)	12.80 um
Final Reaction Forc	433.47 N (at 20 um)	Solving (Active)	38.0 N (at 20 um)
Softening Load Dro	15.75%	Solving (Active)	73.88%
Terminal Damage d_m	1.000000 (Full)	0.0528 (at 3.87 um)	1.000 (Broken)
Crack Chord Angle	-57.95 deg	Active (Solving)	-49.30 deg
Bottom Exit (y=0)	x = 0.813 mm	Active (Solving)	x = 0.930 mm
Physical Solver Stat	COMPLETED (Exit 0)	RUNNING (0 Cutbacks	Published Baseline